

- 6 The points A, B, C have position vectors

$$\mathbf{i} + \mathbf{j}, \quad -\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}, \quad -2\mathbf{i} + \mathbf{j} + 3\mathbf{k},$$

respectively, relative to the origin O .

- (a) Find the equation of the plane ABC , giving your answer in the form $ax + by + cz = d$. [5]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Find the perpendicular distance from O to the plane ABC . [2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) Find a vector equation of the common perpendicular to the lines OC and AB .

[8]

[illegible]